Forcing in K (eccentric anomaly)

$$\dot{p}_i = \ldots$$

Trigonometric polynomials

$$\sin^j K \quad \cos^k K \quad \cdot \quad P_n(\cos\theta)$$

$$(1 - g\cos G)^k$$

$k = 1, 2, 3, \ldots$ from Kepler's equation

$$\frac{a}{r} = \frac{1}{1 - p_1 \sin K - p_2 \cos K}$$

$$\left(\frac{a}{r}\right)^3 \sin^2\theta = \frac{(\ldots)^2}{(1 - g\cos G)^3}$$

Rational–trigonometric:
nested denominators,
no closed-form orbit average

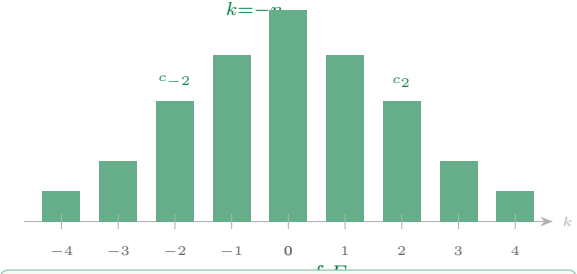
✗

$K \to F$

change of variable

Forcing in F (complex true anomaly)

$$\sum_{k=-n}^{n} c_k F^k$$



Substitution rules:

$$\sin u = \frac{wF - (wF)^{-1}}{2i}$$

$$\cos u = \frac{wF + (wF)^{-1}}{2}$$

Orbit average = c_0 coefficient

(constant term of the Laurent polynomial)

$$\frac{1}{2\pi} \oint f(F) \frac{dM}{dF}\, dF = c_0$$

✓

Same physics, radically simpler algebra.

The Möbius substitution $F = (z - q)/(1 - qz)$ converts all denominators $(1 - g\cos G)^k$ into polynomial prefactors absorbed by dM/dF.